A Data-Driven Forecasting and Inventory Optimization Framework for Apparel Retail Under Sparse-Data Conditions

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In

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Submitted by

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I hereby certify that the work which is being presented in this report entitled "*A Data-Driven Forecasting and Inventory Optimization Framework for Apparel Retail Under Sparse-Data Conditions"*, in fulfilment of the requirements for the award of the MASTER OF TECHNOLOGY in OPERATIONS RESEARCH and submitted to the Department of Management, National Institute of Technology Durgapur, is an authentic record of my own work carried out by me during July 2025 to April 2026, under the guidance of Dr. Goutam Panigrahi, NIT Durgapur. The matter presented in this report has not been submitted by me for the award of any other degree of this or any other Institute.

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Abstract

This thesis introduces a data-driven methodology for forecasting product demand and optimizing inventory management within an apparel retail context. The source dataset presented inherent constraints — specifically, sparse transactional records with insufficient historical depth and an absence of critical explanatory variables such as trend indicators, weather conditions, and product-level attributes.

Such constraints are characteristic of small to medium-sized retail businesses, where data infrastructure remains underdeveloped and archival Records tend to be fragmented or inconsistently maintained.

To overcome these limitations, Google Trends data was integrated as a supplementary signal for consumer demand, enabling aggregated sales figures to be temporally redistributed using trend-derived weighting — thereby reconstructing a statistically coherent time series.

A suite of feature engineering methods was applied, encompassing lagged variables, rolling window averages, and seasonal dummy indicators. Reference models were established to serve as performance benchmarks, and **XGBoost** was ultimately adopted as the core forecasting engine based on its demonstrated predictive superiority.

Forecast uncertainty was quantified through residual-based analysis, and a tail-risk methodology was employed to calculate safety stock levels. A simulation-driven inventory framework was subsequently constructed to assess system performance across cost efficiency and service-level dimensions.

Collectively, this work illustrates a structured progression from theoretical design a fully operational, decision-support inventory optimization system.

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Symbols & Abbreviations

SYMBOLS

Symbol	Description
DEMAND & FORECASTING	
d_t	Actual demand at time t
$\hat{d}_t$	Predicted (forecast) demand
e_t	Forecast error at time t
n	Number of observations
$\bar{d}$	Average demand
σ_d	Standard deviation of demand
w_t	Time-series weight at time t
g_t	Google Trends index value at time t
INVENTORY & REPLENISHMENT	
L	Lead time
SS	Safety stock
R	Reorder point
Q	Order quantity
S	Base-stock level
D	Total demand
I_t	Inventory level at time t
B_t	Stockout quantity at time t
COST PARAMETERS	
K	Ordering cost
h	Holding cost
h_u	Holding cost per unit
b	Stockout cost per unit
TC	Total cost

ABBREVIATIONS

Term	Full Form
FORECASTING & TIME SERIES	
ARIMA	AutoRegressive Integrated Moving Average
SARIMA	Seasonal AutoRegressive Integrated Moving Average
MACHINE LEARNING MODELS	
LSTM	Long Short-Term Memory
XGBoost	Extreme Gradient Boosting
SVR	Support Vector Regression
ELM	Extreme Learning Machine
KM	K-Means Clustering
ML	Machine Learning
AI	Artificial Intelligence
PERFORMANCE METRICS	
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
INVENTORY MANAGEMENT	
EOQ	Economic Order Quantity
SKU	Stock Keeping Unit
MCS	Monte Carlo Simulation

Chapter 1

Introduction

1.1 Background and Motivation

The apparel retail sector operates within one of the most dynamically volatile demand environments across all consumer industries. Unlike commodity goods, Clothing and fashion products are subject to rapid shifts in consumer demand preference, pronounced seasonal cycles, and sensitivity to external influences ranging from emerging style trends to macroeconomic conditions. These characteristics make demand forecasting an inherently complex under the consequences of forecasting error — whether manifested as excess inventory or stockout events — carry significant financial and operational implications.

Inventory management, in this context, extends well beyond mere stock replenishment. It constitutes a strategic function that directly influences customer satisfaction, working capital efficiency, and long-term profitability. Overstocking results in markdowns, increased holding costs, and capital inefficiency, while understocking leads to lost revenue and diminished consumer trust. The ability to anticipate demand with reasonable accuracy is therefore not a peripheral capability but a central operational requirement.

Recent advances in machine learning and data-driven forecasting have opened new avenues for improving predictive accuracy, even in settings where data quality and availability are imperfect. This thesis investigates how such methods can be systematically applied to a real-world apparel retail environment to construct a robust, decision-oriented inventory optimization framework — one capable of operating effectively despite the constraints imposed by limited historical data.

1.2 Data Availability Challenges in Retail Forecasting

A foundational prerequisite for any reliable forecasting system is the availability of structured, high-quality historical data. However, in practice — particularly within small and medium-sized retail enterprises operating in developing and emerging markets — this prerequisite is rarely met in full. Data collection infrastructure in such environments is typically designed to serve

immediate transactional purposes, such as point-of-sale processing and basic accounting, rather than to support downstream analytical or predictive use cases.

As a result, the datasets generated by these systems tend to exhibit several structural limitations. Historical depth is often insufficient to capture meaningful seasonal cycles or long-run demand patterns. Feature representation is inconsistent, with critical explanatory variables either entirely absent or recorded only sporadically. Variables that are well established in the forecasting literature as significant demand drivers — including promotional activity, localized weather conditions, competitive pricing signals, and granular product-level attributes — are rarely available in usable form.

In the context of this project, the operational dataset comprised approximately one year of transaction-level sales records with a constrained set of attributes. Key demand-influencing variables such as marketing expenditure, discount schedules, and product metadata were either missing or could not be reliably reconstructed from available records. Furthermore, the temporal coverage of the dataset was insufficient to support direct estimation of seasonal demand components using conventional time-series decomposition techniques.

These limitations are not merely inconveniences; they fundamentally constrain the class of models that can be responsibly applied. Traditional forecasting approaches — including ARIMA-family models and exponential smoothing frameworks — are predicated on the assumption of sufficient historical observations and well-defined periodic structures. When these assumptions cannot be satisfied, alternative strategies for data enrichment and signal augmentation become necessary preconditions for meaningful forecasting.

This thesis treats data scarcity not as an obstacle to be acknowledged and set aside, but as a central methodological challenge to be addressed through principled engineering and the integration of externally available information sources.

1.3 Problem Statement

The central problem addressed in this thesis is the construction of an accurate, operational demand forecasting and inventory optimization system under conditions of limited, incomplete, and structurally constrained retail data.

Specifically, the problem is characterised by the following interrelated challenges:

i. **Insufficient historical depth:** The available sales record spans a single calendar year, which is inadequate for directly estimating seasonal demand patterns or long-term trend components with statistical confidence.

ii. **Absent explanatory variables:** Demand-relevant features including promotional schedules, weather data, and product-level metadata are either unavailable or too sparsely recorded to be incorporated directly into a supervised learning framework.

iii. **Forecasting under uncertainty:** The inherent variability of apparel demand — compounded by data limitations — necessitates an explicit treatment of forecast uncertainty,

particularly in the tails of the demand distribution, which are most consequential for inventory decision-making.

iv. **Inventory policy design:** Translating probabilistic demand forecasts into actionable, cost-efficient inventory replenishment policies requires a structured decision framework that balances service-level objectives against holding and stockout costs.

Addressing these challenges in an integrated and rigorous manner constitutes the primary scope of this research.

1.4 Research Objectives

This thesis pursues the following interconnected research objectives:

1. **Data Enrichment and Time-Series Reconstruction:**
   To augment the constrained operational dataset by incorporating externally sourced demand signals — specifically, Google Trends data — and to use these signals to reconstruct a temporally meaningful and analytically suitable time series.

2. **Feature Engineering and Representation:**
   To derive a comprehensive set of predictive features from the reconstructed time series, including lag-based temporal variables, rolling statistical aggregates, and calendar-driven seasonal indicators.

3. **Forecasting Model Development and Selection:**
   To develop and benchmark multiple demand forecasting models — ranging from statistical baselines to machine learning approaches — and to identify the best-performing model based on systematic evaluation criteria.

4. **Uncertainty Quantification:**
   To model forecast residuals in order to characterise predictive uncertainty, with particular attention to tail-risk behaviour that governs safety stock determination under non-standard demand distributions.

5. **Inventory Policy Design:**
   To translate demand forecasts and uncertainty estimates into a structured inventory replenishment policy, incorporating safety stock and reorder point calculations grounded in risk-adjusted service-level targets.

6. **Simulation-Based Performance Evaluation:**
   To validate the end-to-end system through a simulation framework that assesses inventory performance using metrics spanning total cost, service level attainment, and stockout frequency.

1.5 Key Contributions

This research makes the following original contributions to the applied domain of retail demand forecasting and inventory management:

- **Trend-Driven Data Reconstruction:** A novel approach is proposed for redistributing aggregated sales data across time using Google Trends as a proportional weighting signal. This enables the construction of a structured, feature-rich time series from sparse and temporally shallow operational records, offering a transferable methodology applicable to other data-scarce retail settings.
- **Feature-Augmented Supervised Forecasting:** A supervised machine learning forecasting pipeline is developed, incorporating engineered temporal features that capture lagged demand dynamics, rolling demand averages, and seasonal periodicity. The framework is designed to maximise predictive signal extraction from limited input data.
- **Tail-Risk Safety Stock Modeling:** Departing from conventional normality assumptions, this thesis introduces a safety stock calculation methodology grounded in empirical residual distributions and tail-risk quantification. This approach more accurately reflects the asymmetric cost structure of inventory decisions in fashion retail.
- **Integrated Simulation-Based Evaluation:** A discrete-event inventory simulation is constructed to evaluate the complete forecasting-to-replenishment pipeline under realistic demand conditions. The simulation provides actionable performance diagnostics across service level, cost, and operational stability dimensions.

1.6 Overview of the Proposed Framework

The remainder of this thesis is organized into six chapters.

Chapter 2 establishes the theoretical and empirical foundation of the study by reviewing existing literature on retail demand forecasting, inventory optimization methodologies, forecast uncertainty, and the integration of external behavioural signals within retail analytics systems.

Chapter 3 develops the methodological architecture of the proposed framework. The chapter formulates the research problem, characterizes the structural limitations of the retail dataset, motivates the integration of external behavioural indicators, and presents the analytical methodology spanning temporal demand reconstruction, feature-engineered forecasting, uncertainty quantification, and inventory policy design.

Chapter 4 presents the engineering implementation of the proposed system. The chapter details the preprocessing pipeline, forecast benchmarking experiments, temporal demand reconstruction workflow, XGBoost forecasting implementation, residual-based uncertainty analysis, inventory parameter estimation, and the simulation-driven operational evaluation framework.

Chapter 5 reports the experimental findings and operational evaluation results. The chapter compares forecasting performance across models, analyzes residual behaviour, evaluates inventory outcomes under different replenishment conditions, and examines the relationship between predictive accuracy and operational inventory performance.

Chapter 6 discusses the broader analytical and practical implications of the study. The chapter interprets the significance of external signal integration, sparse-data forecasting, uncertainty-aware inventory planning, and simulation-driven decision support within apparel retail environments.

Finally, Chapter 7 concludes the thesis by summarizing the primary contributions, highlighting methodological limitations, and outlining potential directions for future research in retail forecasting and inventory analytics.

Chapter 2

Literature Review

2.1 Introduction

Demand forecasting and inventory management are critical components of retail supply chain analytics, particularly in the apparel industry where demand patterns are highly volatile. The fashion retail sector is characterized by short product life cycles, rapid changes in consumer preferences, and limited historical data, making accurate forecasting a challenging task.

This chapter reviews existing literature across three key domains: (i) demand forecasting in fashion retail, (ii) machine learning approaches for forecasting, and (iii) inventory management under stochastic demand.

2.2 Demand Forecasting in Fashion Retail

Vashishtha et al. [1] highlight that fashion demand is inherently difficult to predict due to sparse and noisy data and lack of historical data.

Similarly, Chen and Lu [2] emphasize that demand forecasting in fast fashion is complicated by uncertain demand and dynamic market conditions.

2.3 Machine Learning Approaches for Demand Forecasting

Chen and Lu [2] propose hybrid forecasting models integrating clustering and machine learning algorithms such as ELM and SVR.

Vashishtha et al. [1] further note that even advanced models may underperform when historical data is limited, highlighting the importance of feature engineering and external data integration.

2.4 Inventory Management under Stochastic Demand

Maitra et al. [3] emphasize that traditional deterministic inventory models are insufficient under stochastic demand conditions.

Their work demonstrates that Monte Carlo simulation provides better risk insights compared to analytical models.

2.5 Integration of Forecasting and Inventory Optimization

Vashishtha et al. [1] show that forecasting should be integrated with inventory planning for better decision-making.

Maitra et al. [3] highlight that simulation-based frameworks allow better evaluation of uncertainty impacts.

2.6 Research Gap

Despite advancements, limitations remain:

- Traditional models fail under limited data
- ML models require large datasets
- Limited use of external signals
- Inventory models assume normal demand
- Lack of integrated frameworks

Chapter 3

Problem Formulation and Methodology

3.1 Introduction

This chapter develops the methodological architecture underpinning the proposed demand forecasting and inventory optimization system. Building upon the theoretical foundations and the research gap identified in Chapter 2, the present chapter articulates the analytical rationale governing the design of each component of this research.

The central methodological challenge arises from a structural mismatch that is common to small and medium-sized retail enterprises: the transactional records available for analysis are temporally shallow, feature-impoverished, and insufficient to satisfy the data requirements of conventional forecasting methods. Rather than treating these limitations as boundary conditions that constrain the analytical scope, this research treats them as primary design constraints that shape a principled and transferable solution strategy.

The chapter proceeds through a logical sequence: it first articulates the formal problem structure, characterises the datasets and their limitations, motivates the choice of external signal integration, presents the comparative model benchmarking rationale, and then develops the full analytical methodology spanning data reconstruction, supervised forecasting, uncertainty quantification, and inventory policy design. Each methodological decision is grounded in the preceding analytical context, ensuring a coherent progression from problem diagnosis to decision-support design.

3.2 Problem Formulation

Demand forecasting in apparel retail occupies an inherently difficult position in the broader landscape of time-series prediction. Unlike commodity markets, where demand follows relatively stable and structurally interpretable patterns, fashion retail demand is shaped by short product life cycles, rapidly shifting consumer preferences, pronounced seasonal cycles, and sensitivity to external trend signals that have no direct representation in transactional records. These domain characteristics impose a high informational burden on any forecasting system

and simultaneously limit the reliability of purely historical extrapolation.

The operational dataset available for this study amplifies these domain-specific difficulties through several interconnected structural limitations.

i. **Constrained historical depth:** Transactional records span approximately one year, precluding direct estimation of seasonal demand cycles through decomposition methods that require multiple years of observations to achieve statistical stability.

ii. **Absent explanatory variables:** Demand-relevant features including promotional activity, pricing schedules, weather conditions, and product-level attributes are either unavailable or inconsistently recorded, removing the explanatory foundation that supervised learning models depend upon.

iii. **Inconsistent product taxonomy:** Product naming conventions across the dataset are non-standardized, complicating direct category-level aggregation and comparison with external information sources.

iv. **Sparse temporal demand structure:** Aggregated observations exhibit limited temporal variability, reducing the discriminative signal available for model training and evaluation.

v. **Compounded forecast uncertainty:** High-volume product categories concentrate a disproportionate share of demand variability, increasing the operational consequences of replenishment errors in precisely those segments where inventory decisions are most consequential.

These limitations collectively motivate a methodology that does not attempt to extract forecasting performance directly from impoverished inputs, but instead transforms the data environment through principled enrichment before applying predictive modeling. The proposed methodology therefore pursues five interrelated objectives: (i) the reconstruction of a temporally coherent demand signal from sparse observations; (ii) the integration of external demand proxies to compensate for absent explanatory variables; (iii) the development of a feature-enriched supervised forecasting model; (iv) the characterisation of forecast uncertainty through empirical residual analysis; and (v) the translation of probabilistic demand estimates into uncertainty-aware inventory replenishment decisions.

3.3 Dataset Description and Analytical Constraints

3.3.1 Primary Retail Sales Dataset

The core dataset comprises SKU-level transactional records obtained from an apparel retailer, covering approximately one year of operations. The dataset contains item-level identifiers, quantities sold, and monetary sales values. Although the records faithfully reflect operational

behaviour, exploratory analysis identified several structural properties that limit their direct analytical utility.

The historical span is insufficient to support reliable seasonal decomposition; conventional time-series approaches typically require a minimum of two to three complete seasonal cycles to estimate periodic structures with statistical confidence. Furthermore, the dataset lacks the explanatory dimensions — promotional calendars, pricing signals, weather variables, and product attributes — that the forecasting literature consistently identifies as primary demand drivers in apparel retail [1, 2]. Table 3.1 summarises the key attributes and constraints of the primary dataset.

Table 3.1: *Structural profile of the primary retail dataset.*

Attribute	Description
Dataset type	Apparel retail transactional records
Historical coverage	Approximately one year
Primary variables	Item name, quantity sold, sales value
Granularity	SKU-level transactions
Structural limitation	Insufficient temporal depth for seasonal estimation
Absent variables	Promotions, pricing, weather, product metadata
Operational challenge	Non-standardized product nomenclature

3.3.2 Benchmarking Dataset

Prior to constructing the operational forecasting pipeline, a comparative evaluation phase was conducted using a separately sourced, feature-rich analytical dataset. This benchmarking dataset offered a richer representational structure, with multiple engineered variables available for model evaluation. Its purpose was not operational deployment but methodological: to compare forecasting architectures under controlled, information-rich conditions in order to identify the most suitable model before applying it to the constraints of the retail dataset. This phased approach ensured that model selection was evidence-based rather than assumption-driven.

3.3.3 Visuelle 2.0 Dataset

Supplementary reference information was drawn from the Visuelle 2.0 dataset, which provides structured fashion retail product attributes and historical retail behaviour. This dataset was incorporated not for direct forecasting purposes but to support category standardization and enrich the conceptual grounding of product classification decisions made during preprocessing.

3.3.4 Google Trends Data

The most consequential supplementary source in this study is the Google Trends dataset. Since the operational dataset lacked sufficient temporal variation, an external proxy for consumer

demand fluctuation was required. Google Trends provides normalized weekly search popularity scores for user-defined query terms, capturing the aggregate information- seeking behaviour of consumers as a reflection of evolving purchasing interest.

The rationale for this choice rests on two empirically supported observations. First, consumer search volume has been demonstrated in the forecasting literature to be a leading indicator of purchasing behaviour, particularly in discretionary product categories such as clothing [2]. Second, the normalized structure of Google Trends data makes it compatible with proportional weighting operations that can redistribute aggregate demand across time without requiring absolute demand figures. This property makes it especially well-suited to the reconstruction methodology described in Section 3.6.

3.4 Comparative Benchmarking and Model Selection Rationale

A recurring challenge in applied forecasting research is the absence of principled model selection: studies frequently adopt a single architecture based on precedent or convenience without demonstrating its empirical advantage over alternatives. To avoid this limitation, this research incorporates an explicit benchmarking phase that evaluates multiple forecasting paradigms under comparable experimental conditions before committing to a primary modeling approach.

The models considered in this phase span the principal methodological traditions present in the demand forecasting literature.

i. **ARIMA:** Represents the canonical approach to linear time-series forecasting, capturing autoregressive and moving-average dependencies within a stationary framework. Its limitations under seasonal and nonlinear demand conditions are well-documented [1].

ii. **SARIMA:** Extends ARIMA with seasonal differencing and seasonal autoregressive terms, enabling explicit modeling of periodic demand structure. It remains constrained, however, to univariate input and linear relationships.

iii. **Prophet:** A decomposable forecasting model designed for trend and seasonality estimation across business time series. While robust to missing data and capable of accommodating multiple seasonalities, it does not leverage engineered external features.

iv. **XGBoost:** A gradient-boosted ensemble learning architecture that operates on an arbitrary feature matrix, enabling nonlinear interactions among lagged demand values, rolling statistics, calendar indicators, and external trend signals to be exploited jointly.

The comparative evaluation demonstrated that XGBoost achieved consistently superior performance across standard accuracy metrics. Its ability to exploit the full feature matrix — incorporating external trend signals, temporal aggregates, and calendar variables simultaneously — produced a richer representation of demand dynamics than any of the univariate or

Table 3.2: *Candidate forecasting models and their principal capabilities.*

Model	Methodological Characteristic
ARIMA	Linear autoregressive forecasting under stationarity
SARIMA	Seasonal extension of ARIMA for periodic demand
Prophet	Decomposable trend-seasonality framework
XGBoost	Nonlinear ensemble learning on engineered feature matrices

linear alternatives. On this empirical basis, XGBoost was adopted as the primary forecasting architecture for the operational phase of this research.

3.5 Overview of the Proposed Analytical Framework

The integrated methodology developed in this study is organized around a two-phase structure, as illustrated in Figure 3.1.

The first phase is methodologically preparatory: it conducts comparative model evaluation using the benchmarking dataset, thereby producing an evidence-based model selection outcome. The second phase is operationally directed: it applies the selected model to the enriched retail dataset, extending the analysis from predictive accuracy into uncertainty quantification and inventory decision support.

This two-phase structure was deliberately chosen to ensure that the final forecasting architecture reflects demonstrated empirical superiority under controlled conditions rather than a priori assumptions. The separation also clarifies the distinct analytical roles of the benchmarking and operational datasets, preventing conflation of the model selection and deployment stages.

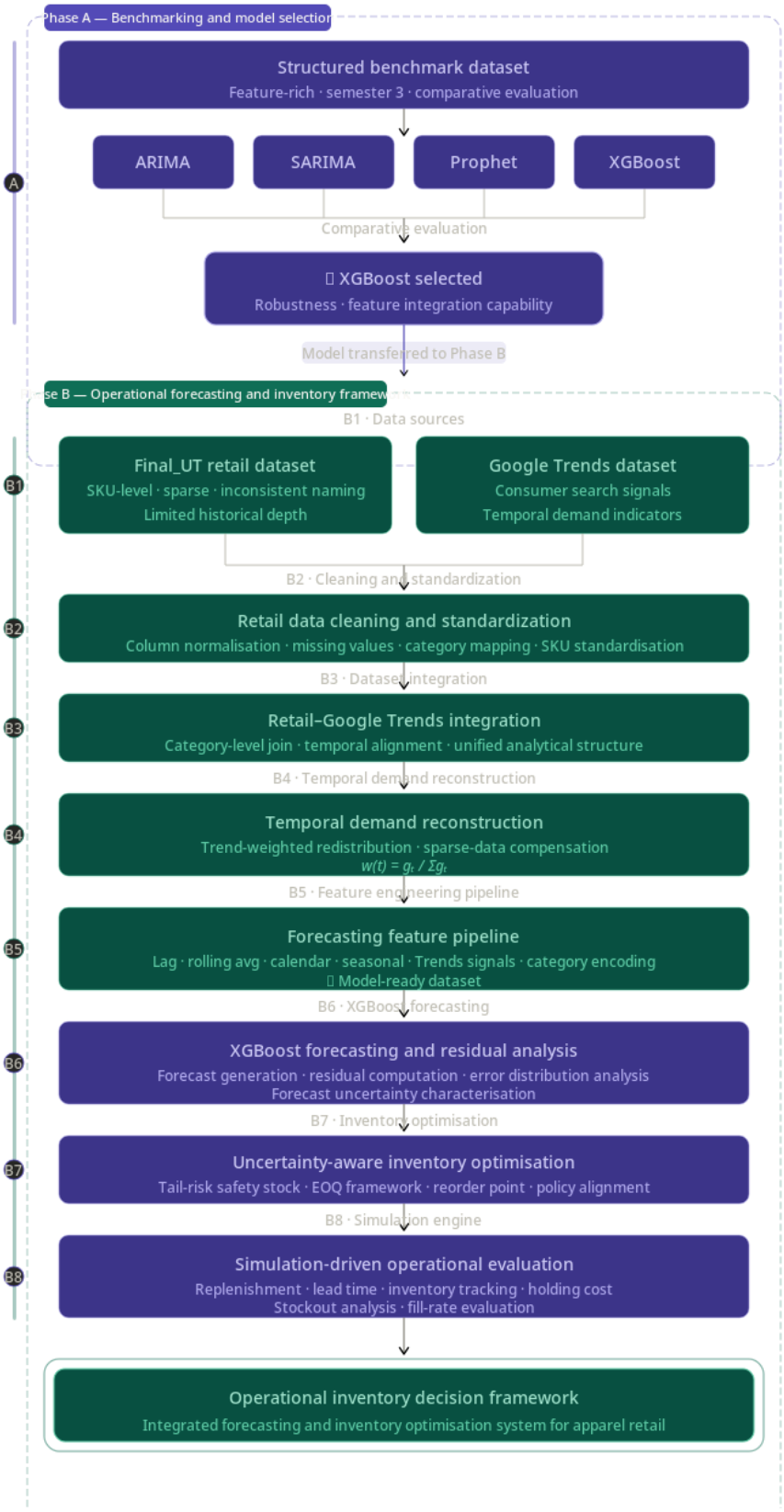


Figure 3.1: *Integrated forecast benchmarking and operational inventory forecasting framework.*

3.6 Temporal Demand Reconstruction Methodology

3.6.1 Analytical Motivation

Standard supervised learning models require a training corpus that captures meaningful variation across the target variable. When the available demand signal is temporally shallow — spanning a single year with limited week-to-week fluctuation — the model's capacity to learn generalizable patterns is fundamentally constrained. Direct application of a feature-based forecasting model to such sparse inputs risks producing estimates that reflect noise rather than structure, with poor generalization to unseen demand conditions.

The reconstruction methodology addresses this by treating the aggregate annual demand as a known total that can be meaningfully redistributed across time using an external proportionality signal. Rather than fabricating demand observations, the approach leverages the information content of Google Trends to assign temporally credible weights to each period within the observation window. The result is a reconstructed time series that respects the observed aggregate while introducing temporal variation grounded in consumer behaviour.

3.6.2 Category Standardization

Meaningful alignment between the retail dataset and Google Trends search categories requires that products be mapped to a common categorical taxonomy. Non-standardized SKU-level names were therefore consolidated into four analytically tractable groupings: long sleeves, short sleeves, lowers, and printed shirts. This aggregation reduces idiosyncratic product-level noise and enables a direct correspondence between retail demand categories and the search query terms used in Google Trends data collection.

Table 3.3: *Illustrative product-to-category mapping used during standardization.*

Raw Product Identifier	Standardized Category
Adidas Black Track TF	Lowers
Graphic Printed Tee	Printed Shirts
Cotton Full Sleeve Top	Long Sleeves
Basic Half Sleeve Tee	Short Sleeves

3.6.3 Trend-Based Temporal Weighting

Once category-level aggregation is established, Google Trends values are normalized within each category to derive temporal weight factors:

$$\square_{\square} = \frac{\square_{\square}}{\sum \square_{\square}} \qquad (3.1)$$

where w_t denotes the temporal weight assigned to period t and g_t is the corresponding Google Trends search popularity score. The normalization in Equation (3.1) ensures that the weights across any given category sum to unity, preserving the aggregate demand total while redistributing it in proportion to observed consumer interest.

This weighting mechanism is conceptually analogous to prior-informed temporal allocation: periods of high search activity receive a larger share of annual demand, reflecting the assumption that search behaviour is correlated with purchasing propensity.

3.6.4 Reconstructed Analytical Dataset

The product of applying these weights to category-level aggregate demand is a reconstructed time series containing, for each observation: date, product category, temporally redistributed demand, and the corresponding Google Trends value. This dataset constitutes the analytical input for all subsequent feature engineering, model training, and inventory analysis stages.

3.7 Forecasting Methodology

3.7.1 Feature Engineering Rationale

The predictive value of any supervised forecasting model is bounded by the informational content of its input features. In the present context, where external demand drivers such as promotions and pricing are unavailable, feature engineering must extract the maximum possible signal from temporal structure and the Google Trends proxy.

The engineered feature set is constructed around four complementary representations:

Lag Features

Lagged demand variables introduce temporal persistence directly into the feature matrix. A one-period lag (Lag-1) captures short-term autocorrelation and provides the model with the most recent observed demand as a conditioning signal. A seven-period lag (Lag-7) encodes weekly periodicity, allowing the model to learn from demand levels observed at the same weekday in the preceding week.

Rolling Window Statistics

Rolling averages over specified lookback windows smooth transient demand fluctuations and expose medium-term demand trends that are not evident from single-period lags. Their inclusion improves the model's ability to distinguish trend-driven demand shifts from idiosyncratic noise.

Calendar Variables

Month and week-of-year indicators capture recurring seasonal structure that would otherwise be invisible to the model in the absence of multi-year training data. These features act as learned seasonal adjustments, enabling the model to modulate predictions in response to calendar position.

External Demand Signal

The contemporaneous Google Trends value is retained as a feature, allowing the model to condition predictions on real-time consumer interest in addition to lagged demand history. This is the primary mechanism through which external information augments the predictive signal.

Table 3.4: *Feature engineering structure and analytical purpose.*

Feature Category	Analytical Purpose
Lag features	Encode temporal demand persistence and periodicity
Rolling statistics	Capture medium-term trend and noise attenuation
Calendar variables	Represent recurring seasonal demand structure
Google Trends signal	Proxy for contemporaneous consumer interest
Category encoding	Distinguish demand dynamics across product groups

3.8 Inventory Optimization Methodology

3.8.1 Uncertainty Quantification Through Residual Analysis

Predictive accuracy, expressed through aggregate error metrics, provides an incomplete basis for inventory decision-making. What matters for replenishment policy design is not the average forecast error but the distribution of that error, and particularly the behaviour of its extremes. A model that achieves low average error but exhibits heavy tails in its residuals may prove more operationally problematic than a less accurate model whose errors are tightly bounded.

Accordingly, this study incorporates residual analysis as a diagnostic and design tool. Forecast residuals are defined as $\varepsilon_t = y_t - \hat{y}_t$ and are examined both for their distributional properties and for systematic patterns across time and product categories. The analysis is designed to identify whether errors exhibit systematic bias, non-stationarity, or category-specific heteroscedasticity.

3.8.2 Safety Stock Estimation

Conventional safety stock formulations apply a service-level multiplier to the standard deviation of demand or forecast error, under the implicit assumption that the error distribution is Gaussian. As discussed in Chapter 2, this assumption is routinely violated in apparel retail, where

demand events in the tails of the distribution are more frequent than a normal model predicts. Underestimating tail risk systematically results in safety stocks that are insufficient to maintain target service levels during high-demand episodes.

To address this, the proposed methodology employs a dual approach: safety stock is estimated both from the standard deviation of residuals and from an empirical tail-risk measure based on a high quantile of the residual distribution. The more conservative of the two estimates is adopted as the operational safety buffer, ensuring protection against both typical variability and extreme deviation events.

The reorder point is subsequently defined as:

$$R = \bar{D} \cdot L + SS \tag{3.2}$$

where $\bar{D}$ denotes average demand per period, L is the replenishment lead time in periods, and SS is the estimated safety stock.

3.8.3 Economic Order Quantity

Cost-efficient replenishment batch sizing is determined through the Economic Order Quantity (EOQ) model, which minimises the combined cost of placing orders and holding inventory:

$$\text{EOQ} = \sqrt{\frac{2DS}{H}} \tag{3.3}$$

where D denotes total demand over the planning horizon, S is the fixed cost per order, and H is the holding cost per unit per period. This formulation provides a category-specific replenishment quantity that balances the trade-off between order frequency and average inventory investment.

3.9 Inventory Simulation Framework

Forecasting accuracy evaluated in isolation captures only the statistical dimension of model performance. To assess operational utility, the study embeds the forecasting pipeline within a simulation-based inventory environment that evaluates the downstream consequences of forecast quality on replenishment decisions.

The simulation implements a base-stock replenishment policy in which orders are triggered when the inventory position falls to or below the estimated reorder point. The order quantity is set to restore the inventory position to the base-stock level, defined as the sum of forecast demand over the lead time and the computed safety stock. The simulation tracks inventory levels, stockout events, fulfillment rates, holding costs, and total inventory expenditure across the evaluation horizon.

This design enables a direct comparison between forecasting strategies in terms of their operational consequences, rather than solely their statistical error properties — a distinction that proves analytically significant in the results presented in Chapter 5.

3.10 Evaluation Methodology

3.10.1 Forecasting Accuracy Metrics

Model predictive performance is quantified through two complementary metrics. Mean Absolute Error (MAE) measures the average magnitude of forecast deviations and is robust to the influence of individual large errors:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{3.4}$$

Root Mean Squared Error (RMSE) penalises large deviations quadratically, making it sensitive to tail events and providing a more stringent accuracy benchmark:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{3.5}$$

Reporting both metrics provides a comprehensive assessment of model performance under average and extreme error conditions respectively.

3.10.2 Operational Inventory Metrics

Inventory system performance is evaluated across five operational dimensions: (i) holding cost, reflecting the cost of maintaining excess inventory; (ii) stockout cost, reflecting the financial and service-level consequences of demand exceeding supply; (iii) total inventory cost, as the aggregate of holding and stockout expenditure; (iv) stockout volume, measuring the total units of unfulfilled demand; and (v) fill rate, defined as the proportion of demand periods satisfied without stockout occurrence. These metrics collectively characterise system performance across both cost efficiency and service reliability dimensions.

3.11 Methodological Cohesion and Chapter Linkage

The methodology presented in this chapter is designed as an integrated system rather than a sequence of independent techniques. The dataset constraints motivate the reconstruction approach; the reconstruction approach produces the analytical inputs necessary for feature engineering; feature engineering enables the supervised forecasting model to operate effectively on sparse data; the forecasting residuals characterise uncertainty in a form that directly informs safety stock design; and the safety stock and EOQ parameters govern the replenishment policy evaluated by the simulation.

This chain of methodological dependencies ensures that each component serves a specific analytical function within the broader system. Chapter 4 demonstrates how this conceptual architecture is operationalized within a concrete software implementation.

3.12 Summary

This chapter developed the methodological rationale for an integrated demand forecasting and inventory optimization system designed under conditions of data scarcity. The structural limitations of the operational dataset were articulated and shown to necessitate the integration of Google Trends as an external demand proxy and the reconstruction of a temporally coherent time series through proportional weighting.

A comparative benchmarking strategy was established to guide evidence-based model selection, leading to the adoption of XGBoost as the primary forecasting architecture. The forecasting methodology was grounded in a principled feature engineering strategy spanning lag-based temporal variables, rolling statistics, calendar indicators, and the external trend signal. Inventory policy design was anchored in empirical residual analysis, departing from conventional distributional assumptions to provide more faithful uncertainty quantification. A simulation-based evaluation protocol was specified to assess the operational consequences of forecasting decisions beyond statistical accuracy metrics alone.

Collectively, these methodological components form a coherent and practically grounded analytical pipeline, the engineering realization of which is described in Chapter 4.

Chapter 4

Implementation and System Development

4.1 Introduction

This chapter documents the engineering execution of the analytical framework specified in Chapter 3. Whereas the preceding chapter established the research rationale and methodological design underlying each system component, the present chapter focuses on how those components were operationalized: the preprocessing pipeline that transformed raw transactional records into a usable analytical structure, the benchmarking experiments that guided model selection, the demand reconstruction workflow that compensated for insufficient temporal depth, the feature engineering and model training pipeline, the residual-based uncertainty analysis, the inventory policy computation, and the simulation engine used for operational evaluation.

The implementation pipeline was constructed as an end-to-end, modular workflow in which each stage processes the outputs of the preceding one, ensuring consistency of data representations and parameter estimates across the full system. Details of the software environment are presented first, followed by a stage-by-stage description of the engineering decisions, code implementations, and resulting analytical structures.

4.2 Development Environment and Software Stack

The system was implemented entirely within the Python ecosystem, which provides comprehensive library support for the machine learning, statistical modeling, and simulation workflows required by this study. All development and experimentation were conducted within the Google Colab environment, enabling iterative prototyping and reproducible execution.

Table 4.1: *Software tools and libraries deployed across the implementation pipeline.*

Tool / Library	Role in Implementation
Python 3	Core implementation language
Pandas	Data preprocessing, transformation, and manipulation
NumPy	Numerical operations and vectorized computation
Scikit-learn	Evaluation metrics and preprocessing utilities
XGBoost	Primary forecasting model
Statsmodels	ARIMA benchmarking implementation
Prophet	Trend-seasonality benchmarking implementation
Matplotlib	Visualization of forecasts, residuals, and inventory dynamics
Google Colab	Interactive development and execution environment

4.3 Retail Data Cleaning and Standardization

The raw transactional dataset exhibited three classes of preprocessing challenges that required systematic resolution before any analytical processing could proceed: non-uniform column naming conventions, inconsistent data types, and non-standardized product identifiers. A multi-stage preprocessing pipeline was implemented to address each class in sequence.

Column names were first normalized to lowercase and stripped of extraneous whitespace, then remapped to semantically consistent identifiers used throughout the downstream pipeline.

```
store_raw.columns = \
    store_raw.columns.str.lower().str.strip()

store_raw = store_raw.rename(columns={
    'item name': 'item',
    'sale quantity': 'quantity',
    'sale amount': 'sales'
})
```

Code 4.1: *Column normalization and renaming.*

Following column standardization, records with missing item identifiers or quantity values were removed, and summary-level rows (such as “Total” entries) that were embedded within the transactional data were excluded to prevent their values from distorting demand aggregations.

```
store_raw = \
    store_raw.dropna(subset=['item', 'quantity'])

store_raw = \
    store_raw[store_raw['item']
    .str.lower() != 'total']
```

Code 4.2: *Removal of incomplete and non-transactional records.*

Product names were subsequently mapped to standardized categories through a rule-based classification function. This step resolves the ambiguity introduced by inconsistent naming conventions and produces the categorical structure required for alignment with Google Trends query terms.

```
def clean_category(x):

    x = str(x).lower().strip()

    if any(word in x for word in
           ['full sleeves', 'jacket', 'hoodie']):
        return 'long sleeve'

    elif any(word in x for word in
             ['pants', 'jeans']):
        return 'lowers'
```

Code 4.3: *Rule-based product category classification.*

The rule-based approach was selected over automated clustering methods given the limited number of well-defined target categories and the interpretability requirement of the resulting labels. The cleaned and categorized dataset formed the input to all subsequent processing stages.

4.4 Forecast Benchmarking Implementation

In accordance with the model selection strategy described in Chapter 3, a comparative benchmarking phase was executed using the feature-rich analytical dataset prior to deploying any model on the sparse retail records. This phase evaluated ARIMA, Prophet, and XGBoost under controlled experimental conditions on a common holdout period.

Each model received the same training and test partitions, produced by a temporal split preserving chronological ordering. The final 20% of observations constituted the evaluation horizon, ensuring that no future information was incorporated into training.

```
split_idx = int(len(lagged) * 0.8)

train = lagged.iloc[:split_idx]
test  = lagged.iloc[split_idx:]
```

Code 4.4: *Temporal train–test partitioning.*

Lag-based input features were constructed for all models that required a supervised tabular format. The following function generated the lagged feature matrix, where each row contains the current target value alongside its preceding □ observations.

```
def create_lagged(df, n_lags):

    data = pd.DataFrame({'y': df.values})

    for i in range(1, n_lags + 1):
        data['lag_' + str(i)] = \
            data['y'].shift(i)

    return data.dropna()
```

Code 4.5: *Lagged feature matrix construction for supervised forecasting.*

The benchmarking results established a clear performance hierarchy: XGBoost achieved the lowest RMSE and MAE on the held-out test set, outperforming both ARIMA and Prophet across all evaluation conditions. This outcome confirmed XGBoost as the appropriate primary architecture for the operational forecasting stage and provided the empirical foundation for its selection.

4.5 Temporal Demand Reconstruction Implementation

The temporal demand reconstruction pipeline constitutes one of the most technically distinctive contributions of this study. Given that the retail dataset provided insufficient temporal variation for direct supervised model training, the pipeline was designed to redistribute known aggregate demand across time using Google Trends signals as an external proportionality mechanism. The implementation proceeded through four sequentially dependent stages.

4.5.1 Google Trends Format Transformation

The raw Google Trends data was structured in a wide format with each apparel category occupying a separate column. To enable category-level join operations with the retail dataset, this structure was converted into a long (tidy) format where each row corresponds to a unique date-category pair.

```
gtrend_long = gtrend.melt(
    id_vars=['date'],
    value_vars=[
        'long sleeves',
        'lowers',
        'short sleeves',
        'printed shirt'
    ],
    var_name='category',
```

```
    value_name='gtrend'
)
```

Code 4.6: *Reshaping Google Trends data from wide to long format.*

The resulting long-format structure placed date and category on equal footing as join keys, enabling precise category-level alignment with the retail demand records.

4.5.2 Dataset Integration

With both datasets organized around a common categorical structure, category-level retail demand totals were computed and merged with the transformed Google Trends records. The merged dataset unified the internal demand signal and the external trend proxy into a single analytical structure, establishing the relational foundation for temporal weighting.

```
df = pd.merge(
    store,
    gtrend_long,
    on='category'
)

df = df[[
    'date',
    'category',
    'sales',
    'gtrend',
    'time_weight'
]]
```

Code 4.7: *Integration of retail demand and Google Trends records by category.*

This merge operation ensured that each retail demand observation was paired with the corresponding Google Trends value for the same product category, enabling subsequent weighting to be applied consistently across the full dataset.

4.5.3 Temporal Weight Computation

Within each product category, Google Trends values were normalized to sum to unity, yielding temporal weight factors that express each period's share of total observed consumer interest. This normalization was applied using a grouped transformation to ensure that weights were computed independently for each category.

```
gtrend_long['time_weight'] = \
    gtrend_long \
    .groupby('category')['gtrend']  \
```

```
    .transform(lambda x: x / x.sum())
```

Code 4.8: *Category-wise normalization of Google Trends values into temporal weights.*

The group-wise normalization guarantees that each category's weights are computed relative to that category's own aggregate search volume, preventing cross-category scale differences from distorting the allocation.

4.5.4 Demand Redistribution

In the final reconstruction stage, the aggregate demand for each category was redistributed across time by multiplying it by the corresponding temporal weight. A scaling factor of 1000 was applied to produce demand values on an operationally interpretable unit scale.

```
df['sales'] = df['sales'] * df['time_weight']

df['sales'] = df['sales'] * 1000
```

Code 4.9: *Trend-proportional redistribution of aggregate category demand.*

The resulting reconstructed time series contains temporally varying demand observations whose aggregate sum matches the observed annual total for each category while exhibiting the week-to-week fluctuation pattern implied by Google Trends. This reconstructed signal constitutes the training data for all subsequent forecasting and inventory analysis. The full reconstruction workflow is summarized in Figure 4.1.

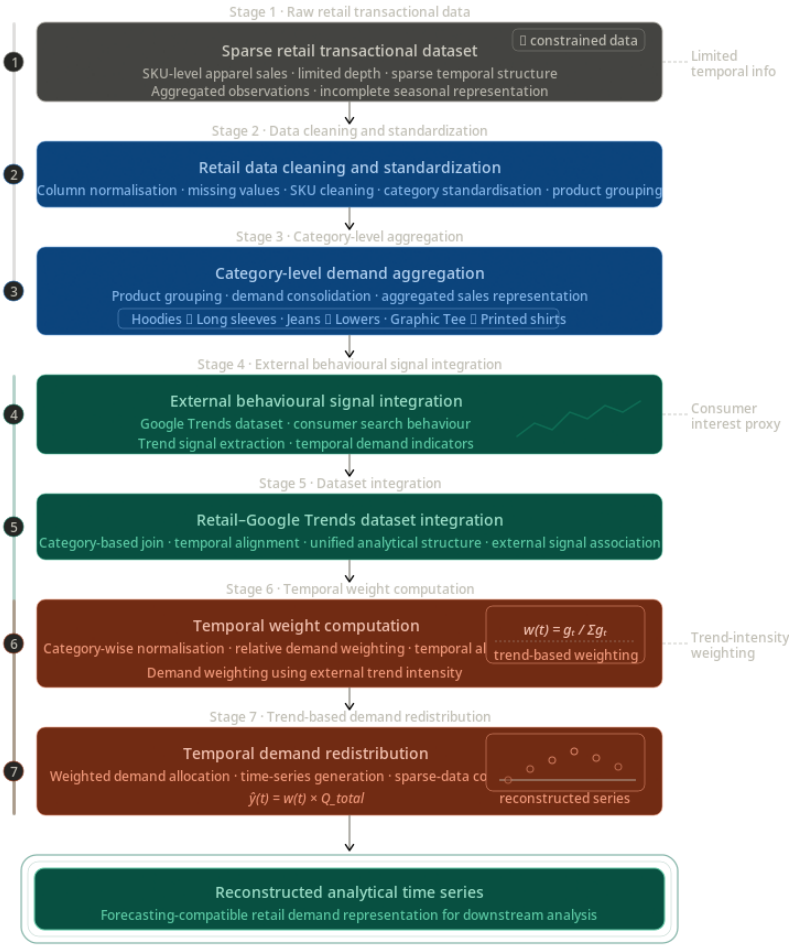


Figure 4.1: *Temporal demand reconstruction pipeline integrating external trend signals with aggregate retail observations.*

4.6 XGBoost Forecasting Implementation

Following the model selection outcome of the benchmarking phase, XGBoost was instantiated and configured for the operational forecasting task. The model was trained on the reconstructed demand dataset using the full set of engineered features described in Chapter 3: lag-1 and lag-7 demand values, rolling window averages, month and week-of-year calendar indicators, the contemporaneous Google Trends signal, and category-level encodings.

The hyperparameter configuration was determined through iterative experimentation, balancing model expressiveness against the risk of overfitting on the reconstructed training data:

```
xgb_model = XGBRegressor(
    n_estimators     = 700,
    learning_rate    = 0.05,
    max_depth        = 6,
    subsample        = 0.8,
    colsample_bytree = 0.8,
```

```
    reg_lambda        = 1.0,
    objective         = 'reg:squarederror',
    random_state      = 42,
    n_jobs            = -1
)
```

Code 4.10: *XGBoost model configuration for demand forecasting.*

The choice of a modest learning rate (0.05) in conjunction with a large number of estimators (700) reflects a slow-learning, high- iteration training strategy that is known to reduce variance and produce more stable ensemble estimates than aggressive learning rate schedules. The subsampling parameters for both observations and features (0.8 each) introduce stochasticity into the tree-building process, further reducing overfitting risk. The L2 regularization term (= 1.0) penalizes leaf weight magnitude, providing additional smoothing of the learned function. Maximum tree depth of 6 was selected to allow the model to capture meaningful nonlinear interactions while preventing excessive depth from memorizing reconstruction artifacts.

```
xgb_model.fit(X_train, y_train)

test_df['pred_xgb'] = \
    xgb_model.predict(X_test)
```

Code 4.11: *Model training and test-set prediction.*

Forecasting performance was evaluated on the held-out test set using RMSE and MAE, providing the quantitative basis for the model comparisons reported in Chapter 5.

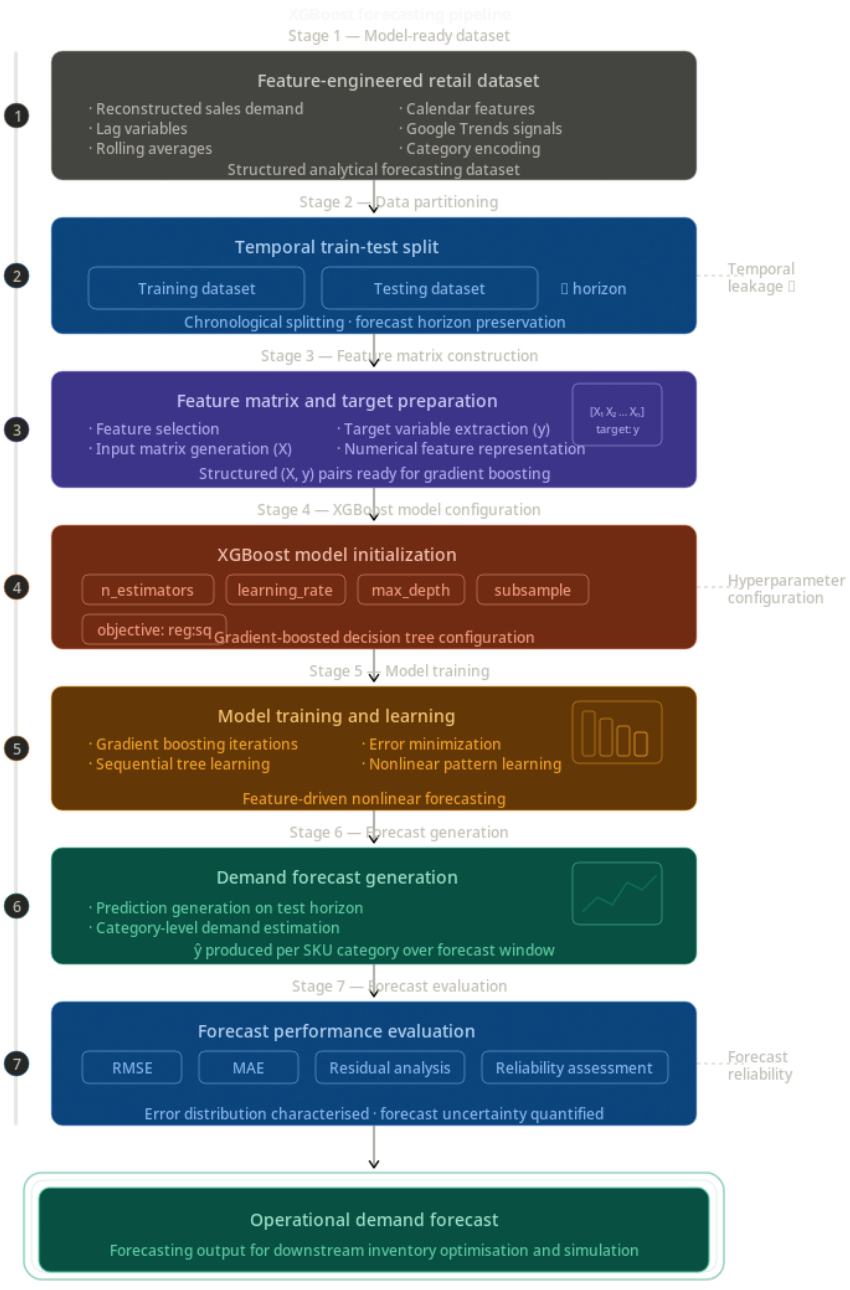


Figure 4.2: *End-to-end implementation workflow of the XGBoost demand forecasting pipeline.*

4.7 Residual Analysis and Uncertainty Modeling

Residual analysis was implemented to characterise the distribution of forecast errors at both aggregate and category levels, providing the empirical foundation for the safety stock calculations described in the following section.

Residuals were computed as the signed difference between observed and predicted demand values:

```
resid_plot_df['resid_xgb'] = (
    resid_plot_df['sales']
    - resid_plot_df['pred_xgb']
)
```

Code 4.12: *Computation of forecast residuals for the XGBoost model.*

The residual series was subsequently analyzed through two complementary lenses. Temporal plots revealed whether systematic bias or non-stationarity was present across the evaluation horizon. Distributional analysis assessed the shape and tail behaviour of the error distribution, with particular attention to deviations from normality that would invalidate standard safety stock assumptions.

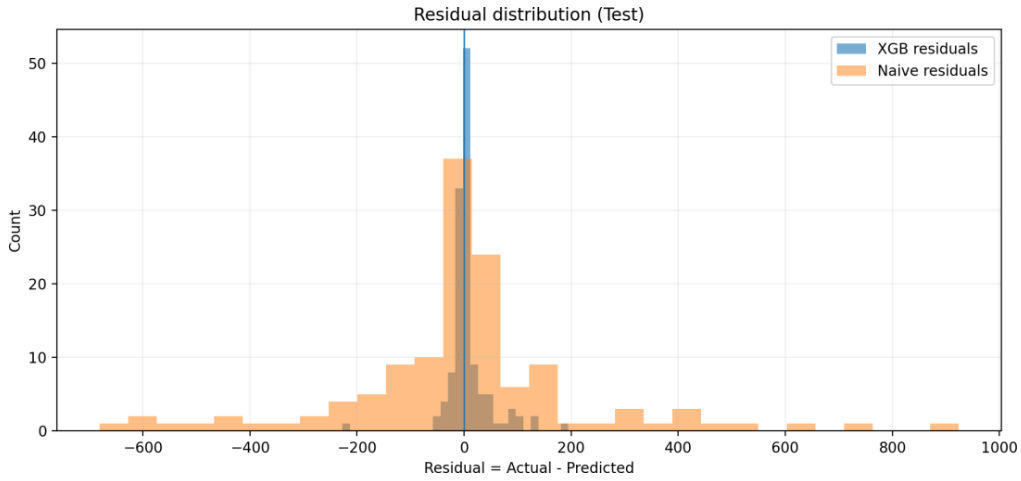


Figure 4.3: *Distributional analysis of XGBoost forecast residuals across product categories.*

The distributional analysis revealed heavier-than-normal tails in the residual distributions, confirming the theoretical concern raised in Chapter 3: that Gaussian safety stock formulas would systematically underestimate extreme demand deviations. To quantify this tail risk empirically, the 95th percentile of the residual distribution was computed for each product category:

```
tail_risk = residuals.quantile(0.95)
```

Code 4.13: *Empirical tail-risk estimation from the residual distribution.*

The resulting tail-risk estimates were subsequently combined with standard deviation-based safety stock values to determine the final safety buffer for each category.

4.8 Inventory Optimization Implementation

The inventory replenishment policy was operationalized by translating forecast outputs and residual uncertainty estimates into three key parameters: safety stock, reorder point, and Economic Order Quantity.

Safety stock for each category was set to the larger of the standard deviation-based estimate and the empirical tail-risk value, providing a conservative buffer calibrated to observed model behaviour:

```
resid_stats['safety_stock'] = np.maximum(
    resid_stats['ss_sigma'],
    resid_stats['tail_risk']
)
```

Code 4.14: *Safety stock determination using dual uncertainty estimates.*

Taking the maximum of the two estimates ensures that the safety buffer accounts for both the overall spread of forecast errors and the severity of extreme deviations, neither of which alone provides a sufficient characterization of inventory risk.

Reorder points were subsequently computed by adding the lead-time demand expectation to the category-specific safety stock:

```
reorder_point = (
    (avg_demand * lead_time)
    + safety_stock
)
```

Code 4.15: *Reorder point computation.*

Economic Order Quantities were computed from annual demand totals, fixed ordering costs, and holding cost rates, implementing the classical EOQ formula from Equation (3.3):

```
EOQ = sqrt(
    (2 * demand * ordering_cost)
    / holding_cost
)
```

Code 4.16: *Economic Order Quantity calculation.*

Together, these three parameters define the replenishment policy supplied to the inventory simulation engine: orders are triggered whenever inventory position reaches the reorder point, and the order quantity is determined by the EOQ calculation.

4.9 Inventory Simulation Engine

To assess how the XGBoost forecasting policy translates into operational outcomes, a discrete-event inventory simulation was constructed and executed over the full test horizon. The sim-

ulation was designed to replicate the sequential nature of inventory management decisions: at each time step, observed demand is fulfilled from available stock; if demand exceeds available inventory, a stockout event is recorded; and replenishment orders are placed according to the base-stock policy when the inventory position falls to or below the computed reorder point.

Demand fulfillment and stockout accounting were implemented as follows:

```
fulfilled_units = min(inv_level, sales_val)

stockout_units = max(0.0,
                     sales_val - inv_level)
```

Code 4.17: *Per-period demand fulfillment and stockout computation.*

Replenishment order quantities were computed at each trigger event by comparing the current inventory position against a base-stock level defined as the sum of forecast demand over the lead time and the estimated safety stock:

```
target    = demand_over_lt + safety_stock

order_qty = max(0.0,
                target - inv_position)
```

Code 4.18: *Inventory replenishment order quantity computation.*

This formulation ensures that replenishment orders are sized to restore the inventory position to its target level, accounting for both anticipated demand during the lead time and the precautionary safety buffer. The simulation recorded holding costs, stockout costs, total costs, fill rates, and stockout volumes at the category level, enabling a comprehensive operational comparison between the XGBoost- driven policy and the naïve baseline.

The simulation framework was not designed to optimize replenishment parameters, but to evaluate the operational consequences of fixed policies derived from the forecasting pipeline. This distinction is important: the simulation serves as a diagnostic tool that surfaces the interaction between forecast signal characteristics and policy assumptions, an interaction whose implications are examined in detail in Chapter 5.

4.10 Integrated System Workflow

The implementation pipeline described in this chapter transforms raw retail transactional data through a sequence of well-defined engineering stages into a decision-support system capable of generating actionable inventory recommendations. The system design was guided by three principles: (i) modularity, such that each stage can be independently validated and modified; (ii) data fidelity, preserving the observed aggregate demand totals throughout reconstruction and feature engineering; and (iii) policy transparency, ensuring that inventory parameters are derived from documented, interpretable computations rather than opaque optimization routines.

The full pipeline — spanning preprocessing, benchmarking, reconstruction, feature engineering, model training, residual analysis, parameter estimation, and simulation — is illustrated in Figure 3.1 from Chapter 3 and implemented as described in the preceding sections.

4.11 Summary

This chapter presented the engineering implementation of the demand forecasting and inventory optimization system. Beginning with raw transactional records, the preprocessing pipeline resolved naming inconsistencies and established a standardized categorical structure. A comparative benchmarking experiment confirmed XGBoost as the superior forecasting architecture under controlled evaluation conditions.

The temporal demand reconstruction pipeline compensated for limited historical depth by integrating Google Trends as an external proportionality signal, producing a temporally coherent demand series from aggregate observations. The reconstructed dataset was enriched with lag-based, rolling-window, calendar, and trend features before being used to train the XGBoost forecasting model under a carefully chosen hyperparameter configuration.

Residual analysis quantified forecast uncertainty at the category level, revealing heavier-than-normal tails that motivated the adoption of an empirical, dual-estimate approach to safety stock calculation. Reorder points and Economic Order Quantities were subsequently derived from these uncertainty estimates, and a discrete-event simulation engine was constructed to evaluate the full forecasting-to-replenishment pipeline under realistic demand conditions.

The results of this operational evaluation, together with their interpretation and practical implications, are presented in the following chapter.

Chapter 5

Results and Discussion

5.1 Introduction

This chapter presents the empirical results obtained from the demand forecasting models and evaluates their effectiveness within an inventory decision-making framework. The analysis is structured in a progressive manner. Section 5.2 provides a comparative evaluation of the forecasting models and reports quantitative accuracy measures. Section 5.3 examines residual behaviour to assess forecast reliability and characterise prediction uncertainty at both aggregate and category levels. Section 5.4 incorporates this uncertainty into inventory parameter estimation, including safety stock, reorder points, and the Economic Order Quantity (EOQ). Section 5.5 embeds the forecasting outputs into an inventory simulation framework and evaluates operational performance in terms of cost efficiency and service level attainment. A consolidated discussion of these findings is offered in Section 5.6, followed by a chapter summary in Section 5.7.

5.2 Forecasting Performance and Model Comparison

5.2.1 Initial Model Comparison

The initial evaluation phase assessed three candidate forecasting approaches — ARIMA, Prophet, and XGBoost — using the analytical dataset constructed in Chapter 3.3. Each model was applied to held-out test data and assessed according to Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

Traditional statistical methods such as ARIMA rely on linear autoregressive and moving average components derived from historical patterns. While capable of representing temporal structure, these models are constrained by stationarity requirements and an inability to incorporate external predictor variables. Prophet addresses some of these limitations through automated decomposition of trend and seasonal components, though it similarly remains restricted to univariate input and cannot leverage the engineered features developed in Chapter 3.7.

XGBoost, by contrast, operates on the full feature matrix comprising lagged demand values, rolling statistics, calendar indicators, and the Google Trends signal. Its gradient-boosted ensemble architecture enables it to capture nonlinear interactions among these predictors, yielding a richer representation of demand dynamics.

Table 5.1: *Initial model accuracy comparison on held-out test data.*

Model	RMSE	MAE
ARIMA	0.01373	0.01082
Prophet	0.00784	0.00644
XGBoost	0.00615	0.00483

As reported in Table 5.1, XGBoost achieves the lowest error across both metrics, with an RMSE of 0.00615 and an MAE of 0.00483. These figures represent improvements of approximately 21.6% and 25.0% over Prophet, and 55.2% and 55.4% over ARIMA respectively. This performance provides a strong empirical basis for adopting XGBoost as the primary forecasting engine.

5.2.2 Post-Reconstruction Evaluation Against Baseline Models

Following dataset reconstruction and the application of the full feature engineering pipeline, XGBoost was benchmarked against three naïve forecasting strategies: the lag-1 naïve model, a seven-period moving average, and the seasonal naïve model based on lag-7. These methods are routinely employed as performance thresholds, as any operational forecasting system must meaningfully surpass them to justify its additional complexity.

Table 5.2: *Post-reconstruction accuracy comparison: XGBoost versus baseline methods.*

Model	RMSE	MAE
XGBoost	44.14	22.76
Moving Average (7-period)	174.44	115.67
Naïve (Lag-1)	228.60	138.31
Seasonal Naïve (Lag-7)	240.77	169.52

Table 5.2 confirms that XGBoost substantially outperforms all baseline methods. Its RMSE of 44.14 is approximately 75% lower than that of the moving average and more than 80% lower than either naïve baseline. The scale of this improvement reflects the degree to which feature-based learning captures demand structure that simple extrapolation rules cannot represent.

Figure 5.1 illustrates the alignment between predicted and observed demand over the test period. XGBoost predictions track actual demand closely, particularly during periods of elevated fluctuation, while the statistical models exhibit greater lag and attenuation in their responses to demand shifts.

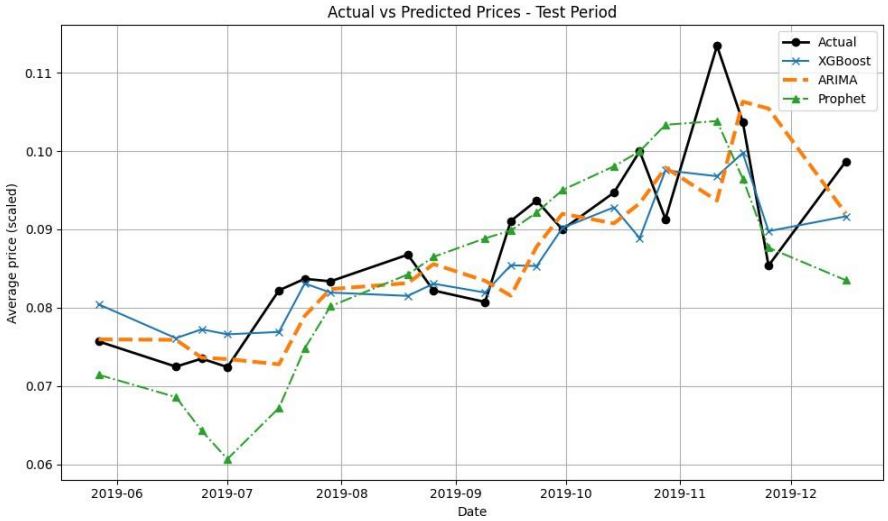


Figure 5.1: *Actual versus predicted demand values: XGBoost, ARIMA, and Prophet.*

5.3 Residual Analysis and Forecast Reliability

Residual analysis extends the evaluation beyond aggregate error metrics by examining the structure and distributional properties of forecast errors. Residuals are defined as the difference between actual and predicted demand values: positive residuals indicate underestimation, while negative residuals reflect overestimation.

5.3.1 Temporal Residual Behaviour

Figure 5.2 displays the temporal evolution of residuals across the evaluation period. XGBoost residuals remain broadly centered around zero throughout, with no systematic directional drift, consistent with an approximately unbiased estimator. The naïve model, by contrast, produces markedly larger oscillations, particularly during periods of rapid demand change where it cannot anticipate shifts in underlying demand structure.

Notwithstanding its overall stability, XGBoost produces isolated spikes of elevated residual magnitude on specific dates. These concentrated errors contribute disproportionately to the RMSE metric, which penalises large deviations quadratically. This property distinguishes RMSE from MAE and explains how a model can achieve strong average accuracy while remaining sensitive to tail events — an observation that directly motivates the tail-risk approach to safety stock estimation in Section 5.4.

A category-level examination reveals further differentiation. The long sleeves category displays largely stable predictions interspersed with occasional large deviations. The lowers category demonstrates consistent, well-balanced residuals with minimal systematic error. The short sleeves category, which exhibits the highest demand volume and intrinsic variability, produces the largest residual magnitudes and the greatest frequency of large individual errors.

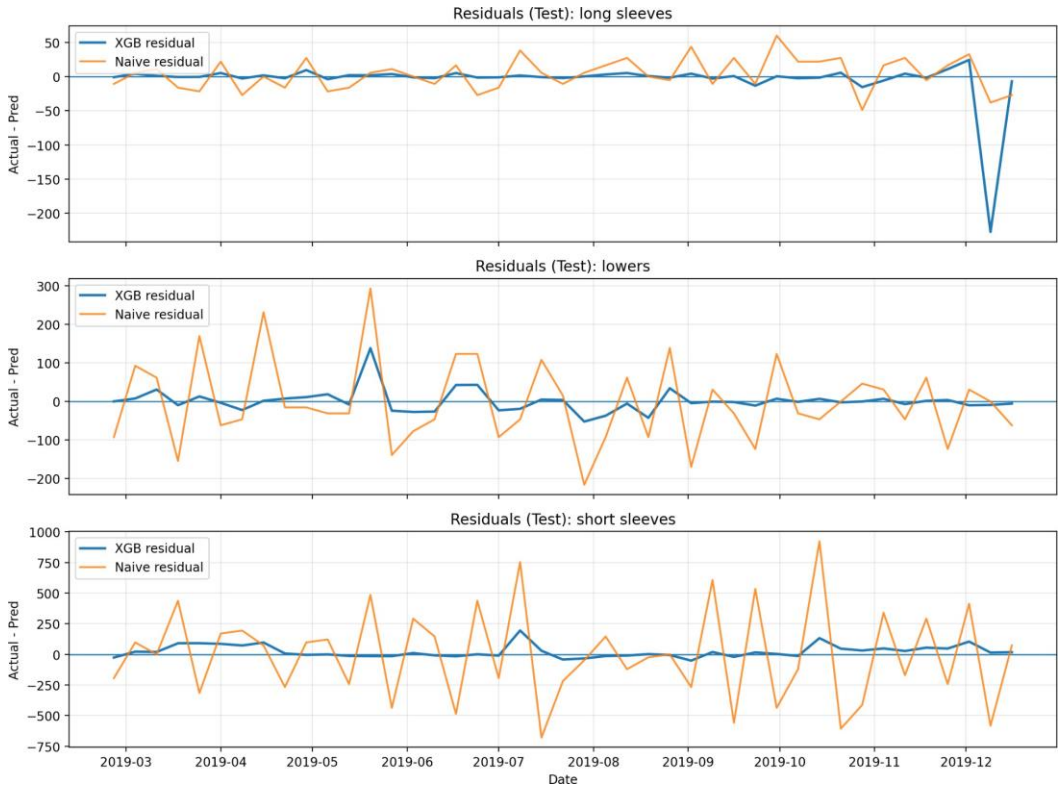


Figure 5.2: *Residual behaviour across product categories for XGBoost and naïve models.*

5.3.2 Residual Distribution

The distributional characteristics of the residuals are presented in Figure 5.3. XGBoost residuals form a concentrated, approximately symmetric distribution centered near zero, consistent with the absence of systematic forecasting bias. The naïve model residuals span a substantially wider range, with pronounced heavy tails reflecting frequent large prediction errors.

Both distributions deviate from strict normality in their tails, indicating that rare but consequential demand deviations occur more frequently than a Gaussian model would predict. This observation reinforces the case for an empirical, non-parametric approach to uncertainty quantification, rather than one grounded in distributional assumptions that understate tail risk.

5.4 Safety Stock and Inventory Parameter Estimation

Forecast uncertainty was incorporated into inventory parameter estimation through a dual methodology combining standard deviation-based and tail-risk-based safety stock calculations. Unlike conventional approaches that assume normally distributed forecast errors, this framework draws directly from the empirical residual distributions characterised in Section 5.3, yielding safety buffers calibrated to observed model behaviour.

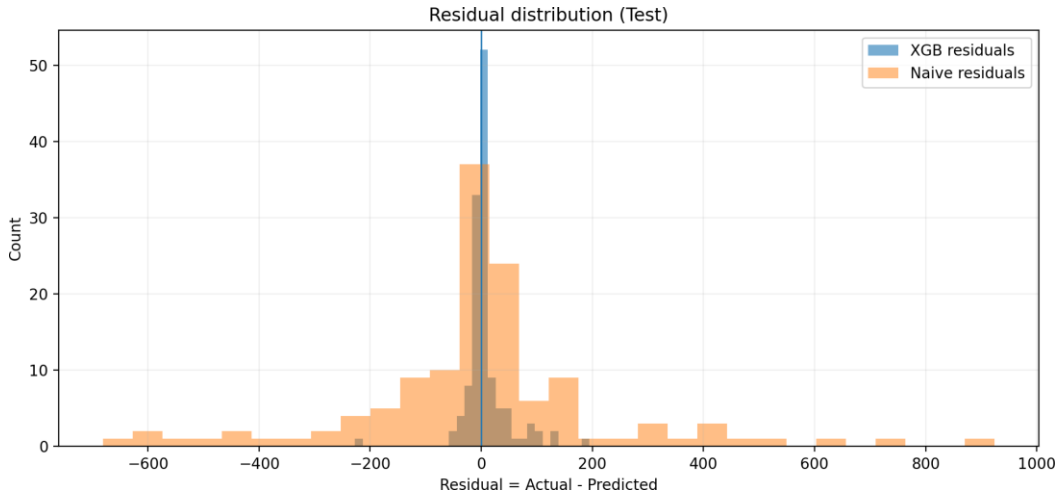


Figure 5.3: *Distribution of residuals for XGBoost and naïve models on the test set.*

5.4.1 Safety Stock and Reorder Point

Safety stock was computed for each product category as a function of residual dispersion and tail-risk exposure. The reorder point was subsequently defined as:

$$R = \bar{d} \cdot L + SS \tag{5.1}$$

where $\bar{d}$ denotes average daily demand, L is the replenishment lead time, and SS is the estimated safety stock.

Table 5.3: *Safety stock, reorder point, and EOQ estimates by product category.*

Category	Avg. Demand		Tail Risk	Safety Stock	Reorder Point	EOQ
Long Sleeves	359.30	37.64	25.86	164.30	2,679.38	2,746.86
Lowers	887.59	31.59	56.94	137.89	6,351.00	4,317.33
Short Sleeves	1550.12	54.12	119.16	236.26	11,087.10	5,705.48

As shown in Table 5.3, safety stock requirements differ markedly across categories. The short sleeves category, which records the highest demand volume and the greatest forecast error variance (σ = 54.12), requires the largest safety buffer of 236.26 units and the highest reorder point of 11,087.10 units. Long sleeves exhibits lower average demand but proportionally elevated variability, producing a moderate safety stock of 164.30 units. Lowers, which demonstrates the most stable demand profile among the three categories, requires the smallest buffer of 137.89 units despite its intermediate demand level. This differentiation across categories illustrates the operational value of grounding safety stock calculations in model-specific residual behaviour rather than applying a uniform rule.

5.4.2 Tail Risk versus Standard Deviation Comparison

To assess the relative contribution of tail risk and overall variance to safety stock sizing, the two measures were compared directly for each category.

Table 5.4: *Comparison of standard deviation-based and tail-risk-based safety stock measures.*

Category	□-Based SS	Tail Risk	Tail/Sigma Ratio	Final Safety Stock
Short Sleeves	236.26	119.16	0.50	236.26
Long Sleeves	164.30	25.86	0.16	164.30
Lowers	137.89	56.94	0.41	137.89

Table 5.4 indicates that the standard deviation-based estimates consistently exceed the empirical tail-risk values across all categories, with tail-to-sigma ratios ranging from 0.16 for long sleeves to 0.50 for short sleeves. The final safety stock adopts the standard deviation-based figure in each case, as this provides the more conservative and operationally robust buffer against demand uncertainty. The tail-risk estimates serve as a complementary diagnostic: they confirm that extreme demand events are less frequent than the standard deviation measure alone might suggest, while remaining material enough — particularly in the short sleeves category — to warrant explicit monitoring in the inventory policy.

5.4.3 Economic Order Quantity

The Economic Order Quantity (EOQ) identifies the replenishment batch size that minimises the combined cost of ordering and inventory holding. It is defined as:

$$\text{EOQ} = \sqrt{\frac{2 \cdot \square \cdot \square}{\square}} \tag{5.2}$$

where □ denotes total demand over the planning horizon, □ is the fixed cost incurred per order, and □ is the holding cost per unit per period. In this study, the ordering cost is set to 500 monetary units and the holding cost to 2 units per item.

The EOQ values reported in Table 5.3 reflect the trade-off between ordering frequency and average inventory level across categories. Short sleeves, with the highest aggregate demand, yields the largest optimal order size of 5,705.48 units, implying that infrequent but large replenishment orders minimise total procurement and holding expenditure for this category. Long sleeves, with its substantially lower demand, is assigned a correspondingly smaller batch of 2,746.86 units. These category-specific EOQ values provide a principled foundation for replenishment scheduling that controls cost across the product range.

5.5 Inventory Simulation and Operational Performance

The forecasting outputs from both the XGBoost model and the naïve baseline were integrated into a base-stock inventory simulation to evaluate their downstream operational impact. Under this policy, a replenishment order is triggered whenever on-hand inventory falls to or below the reorder point, with the order quantity restoring stock to the base-stock level. Performance was assessed across five metrics: holding cost, stockout cost, total cost, stockout volume, and fill rate.

5.5.1 Overall Performance Comparison

Table 5.5: *Inventory simulation results: XGBoost versus naïve baseline under a base-stock policy.*

Model	Holding Cost	Stockout Cost	Total Cost	Stockouts	Fill Rate
XGBoost	56,434.91	24,089.88	80,524.79	4,817.98	0.9589
Naïve	53,057.64	17,779.92	70,837.57	3,555.98	0.9697

Table 5.5 presents a result that merits careful interpretation: the naïve model achieves a lower total cost (70,837.57 versus 80,524.79) and a higher fill rate (0.9697 versus 0.9589) than XGBoost under the base-stock policy. This finding appears counterintuitive given the substantially superior forecasting accuracy demonstrated in Section 5.2, and is examined in detail below.

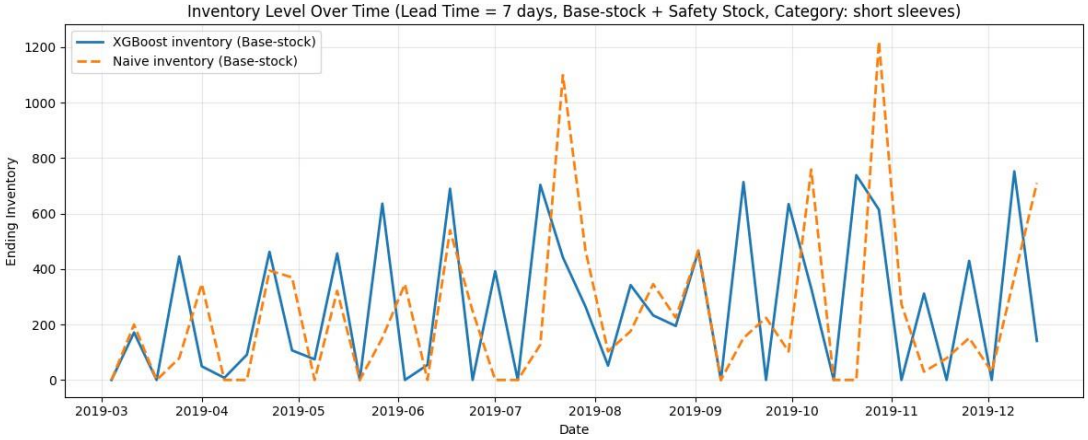


Figure 5.4: *Simulated inventory level trajectories for the short sleeves category: XGBoost policy versus naïve baseline.*

Figure 5.4 illustrates the inventory dynamics of the short sleeves category — the most demanding and variable product group — over the simulation horizon. The XGBoost-driven policy produces more frequent and sharper adjustments in inventory levels, whereas the naïve policy generates smoother, more gradual trajectories that interact more predictably with the fixed replenishment rules of the base-stock framework.

5.5.2 Interpretation of Simulation Results

The divergence between statistical forecasting performance and inventory simulation outcomes reflects a fundamental tension between model expressiveness and policy alignment. Four interacting mechanisms account for the observed pattern.

Forecast Responsiveness and Replenishment Variability. XGBoost generates predictions that adapt dynamically to incoming feature signals, including trend movements and lagged demand patterns. This responsiveness, while beneficial for predictive accuracy, produces more variable replenishment signals under a fixed base-stock policy. The resulting variability in order quantities amplifies inventory fluctuations, increasing both holding costs and stockout exposure relative to what a smoother demand signal would induce.

Stability of the Naïve Forecast Signal. The naïve model produces forecasts that are inherently smooth, as each prediction is simply the most recently observed demand value. Although this approach sacrifices accuracy, its stability aligns well with the assumptions embedded in the base-stock policy, yielding more consistent replenishment decisions and lower aggregate costs under the simulation conditions applied.

Fixed Safety Stock as a Policy Constraint. The base-stock simulation applies a fixed safety stock level derived from historical residuals. This static buffer is well-matched to the predictable error profile of the naïve model, but insufficient to absorb the more variable and occasionally extreme errors produced by XGBoost in high-demand categories. A dynamic safety stock mechanism that adjusts to real-time forecast uncertainty would more appropriately accommodate the output characteristics of an expressive model.

Category-Level Sensitivity. The short sleeves category, which accounts for the majority of total demand and exhibits the greatest residual variability, is the primary driver of the cost differential. Occasional large XGBoost residuals in this category, as identified in Section 5.3, translate directly into elevated stockout events under the fixed-policy simulation, increasing both stockout cost and total cost relative to the naïve baseline.

Key Insight. A critical implication of these findings is that superior forecasting accuracy does not automatically translate into superior inventory performance. The operational effectiveness of a forecasting model depends not only on its predictive capability but on its alignment with the inventory replenishment policy in which it operates. Advanced models such as XGBoost require adaptive, uncertainty-aware inventory policies — such as dynamic reorder points or rolling safety stock recalculation — to fully realise the value of their predictive accuracy in operational settings.

5.6 Discussion

The results presented in this chapter collectively illuminate both the capabilities and the boundary conditions of the integrated forecasting and inventory framework developed in this thesis. Several overarching observations merit explicit discussion.

Predictive Superiority of Feature-Augmented Machine Learning. XGBoost consistently achieves lower prediction errors than ARIMA, Prophet, and all naïve baseline methods across both evaluation phases. The magnitude of the performance differential — particularly against the naïve and seasonal naïve baselines in the post-reconstruction evaluation — confirms that engineered temporal features and external demand signals provide substantive informational value. This finding is consistent with the broader literature on machine learning for demand forecasting, which identifies feature richness and nonlinear modeling capacity as primary sources of predictive advantage in retail settings characterised by complex, interacting demand drivers.

Non-Normality of Forecast Errors and Inventory Implications. Residual analysis reveals that forecast errors from both XGBoost and the naïve model exhibit heavier tails than a normal distribution would predict. This finding carries direct consequences for safety stock design. Conventional safety stock formulas that apply a service-level multiplier to the standard deviation of demand implicitly assume Gaussian residuals; under this assumption, the probability of extreme demand events is systematically underestimated. The empirical tail-risk approach adopted in this study provides a more faithful representation of error behaviour, particularly in high-variability categories where the consequences of underestimating tail risk are most operationally significant.

The Accuracy–Policy Alignment Gap. The inventory simulation results expose a gap between forecasting accuracy and operational performance that is frequently overlooked when forecasting models are evaluated in isolation. The naïve model's superior simulation outcomes — despite its substantially inferior predictive accuracy — demonstrate that policy compatibility is as consequential as forecast quality. The base-stock policy evaluated here was designed under assumptions of moderate, stable demand signals; the more expressive output of XGBoost is not well matched to these assumptions without complementary adaptations in the policy structure. This finding underscores the importance of jointly designing forecasting models and inventory policies, rather than treating them as independent optimization problems.

Practical Implications for Retail Inventory Management. The findings carry tangible implications for practitioners in apparel retail and comparable data-scarce environments. Machine learning forecasting approaches can meaningfully improve demand prediction even when historical data is limited, provided that external signals and principled feature engineering are employed. Safety stock calculations should be grounded in empirical residual analysis rather

than parametric distributional assumptions, as the latter may underprotect against the tail events that drive stockout costs. Furthermore, the deployment of advanced forecasting models should be accompanied by a review of the inventory policy framework to ensure that the informational value of improved predictions is captured in operational decisions rather than absorbed by policy rigidity.

5.7 Summary

This chapter provided a comprehensive empirical evaluation of the demand forecasting and inventory optimization framework developed in this thesis. The principal findings are as follows:

- XGBoost achieved the lowest prediction error among all evaluated models in both the initial comparison and the post-reconstruction baseline benchmarking, with RMSE improvements exceeding 75% over the best-performing naïve method, confirming its suitability as the primary forecasting engine.
- Residual analysis confirmed that XGBoost forecasts are approximately unbiased across the test period, with substantially lower error variance than the naïve baseline. Isolated tail events, particularly in the short sleeves category, account for a disproportionate share of RMSE and motivate empirical approaches to uncertainty quantification.
- Safety stock, reorder point, and EOQ parameters were derived on a per-category basis using both standard deviation-based and tail-risk-based methods. The short sleeves category required the largest safety buffer and replenishment batch, consistent with its higher demand volume and forecast error variability.
- Inventory simulation revealed that the naïve model achieves lower total cost (70,837.57 versus 80,524.79) and a higher fill rate (0.9697 versus 0.9589) than XGBoost under a fixed base-stock policy, a result attributable to the mismatch between the dynamic forecast signal of XGBoost and the static assumptions of the base-stock framework.
- The central practical implication is that forecasting accuracy and inventory performance are jointly determined by both model capability and policy alignment; advanced forecasting models require adaptive inventory policies to fully realise their operational benefits.

Chapter 6

Practical Implications for Retail Inventory Planning

6.1 Introduction

The preceding chapter established the empirical performance of the forecasting models and evaluated their integration within an inventory decision-making framework. While Chapter 5 focused on quantitative evaluation and simulation-based analysis, the present chapter interprets these findings from an operational and managerial perspective.

The objective of this chapter is to translate the analytical results into actionable insights for retail inventory planning. Particular emphasis is placed on the interaction between forecasting accuracy, uncertainty characterisation, and inventory policy design. The discussion proceeds by examining implications for forecasting practice, inventory control, risk management, and strategic retail decision-making.

6.2 Implications for Demand Forecasting

The results presented in Chapter 5 demonstrate that feature-augmented machine learning models are capable of capturing complex demand dynamics more effectively than traditional time-series approaches. The superior performance of XGBoost reflects its ability to incorporate multiple sources of information, including lagged demand, rolling statistics, calendar effects, and external trend signals.

From a practical perspective, this finding suggests that retailers can substantially improve demand visibility by adopting forecasting frameworks that extend beyond purely historical extrapolation. In particular, the integration of external signals such as search trends provides a mechanism for capturing shifts in consumer interest that are not observable in transactional data alone.

However, the results also indicate that forecasting accuracy, when assessed in isolation, provides only a partial measure of model effectiveness. The value of a forecasting system

is ultimately determined by its contribution to downstream decision processes, particularly inventory planning and replenishment. This observation underscores the need to evaluate forecasting models within an integrated decision-making context rather than as standalone predictive tools.

6.3 Implications for Inventory Planning and Replenishment

The integration of forecasting outputs into the inventory simulation framework reveals the critical importance of alignment between forecasting behaviour and inventory policy design. The base-stock policy employed in this study assumes a relatively stable demand signal, with replenishment decisions driven by fixed reorder thresholds and safety stock levels.

Under this policy structure, the naïve forecasting model — despite its inferior predictive accuracy — produces smoother demand estimates that are well aligned with the policy assumptions. This alignment results in more stable inventory trajectories and improved cost performance within the simulation environment.

By contrast, the XGBoost model generates forecasts that respond dynamically to changes in demand drivers. While this responsiveness enhances predictive accuracy, it introduces variability into the replenishment process when combined with a fixed inventory policy. The resulting mismatch leads to increased fluctuations in inventory levels and higher exposure to stockout events.

This finding carries a clear practical implication: inventory policies should not be designed independently of forecasting models. Instead, forecasting and replenishment must be treated as a coupled system in which improvements in predictive capability are matched by corresponding adaptations in policy structure.

6.4 Safety Stock and Risk Management

The residual-based uncertainty modeling framework developed in this study provides a more realistic representation of demand variability than conventional approaches based on distributional assumptions. By analysing the empirical behaviour of forecast errors, it becomes possible to characterise both typical variability and extreme deviations in demand.

The results indicate that safety stock requirements differ substantially across product categories, reflecting variations in both demand magnitude and forecast uncertainty. High-demand categories with greater variability require larger safety buffers to maintain service levels, while more stable categories can operate with comparatively lower inventory reserves.

From a managerial standpoint, this highlights the importance of category-specific risk management strategies. Uniform safety stock policies may lead to inefficiencies, either through excessive holding costs or insufficient protection against stockouts. Incorporating empirical residual analysis into safety stock determination enables more precise calibration of inventory buffers, improving both cost efficiency and service performance.

6.5 Cost Efficiency and Service Level Trade-offs

The simulation results reveal a fundamental trade-off between predictive accuracy and operational performance. While advanced forecasting models reduce average prediction error, their behaviour may introduce variability that affects inventory decisions under fixed policy constraints.

Simpler forecasting approaches, although less accurate in a statistical sense, may provide more stable signals that interact more effectively with traditional inventory control mechanisms. This stability can lead to improved cost outcomes and higher service levels within certain policy frameworks.

For practitioners, this implies that model selection should not be based solely on accuracy metrics such as RMSE or MAE. Instead, forecasting models should be evaluated in terms of their impact on operational objectives, including cost minimisation and service level attainment.

6.6 Category-Level Decision Insights

The analysis further demonstrates that demand behaviour varies significantly across product categories, with important implications for inventory management. High-volume categories, such as short sleeves, exhibit greater variability and are more sensitive to forecasting errors, making them critical drivers of overall system performance.

In contrast, categories with more stable demand patterns allow for tighter inventory control and reduced safety stock requirements. These differences suggest that a uniform inventory policy may not be appropriate across all product groups.

Retailers should therefore adopt differentiated strategies that account for category-specific demand characteristics, uncertainty levels, and service requirements. Such an approach enables more efficient allocation of inventory resources and improves overall system robustness.

6.7 Strategic Implications for Retail Operations

Beyond immediate operational considerations, the findings of this study have broader strategic implications for retail organisations. The integration of forecasting, uncertainty modeling, and inventory optimization supports a transition from reactive decision-making to a more proactive and data-driven approach.

By leveraging advanced analytics, retailers can improve coordination between demand planning, procurement, and supply chain operations. This integration enhances the ability to anticipate demand fluctuations, allocate resources efficiently, and respond to market changes in a timely manner.

Furthermore, the methodology developed in this thesis is scalable and can be extended to larger retail environments with more complex product assortments and supply chain structures.

6.8 Limitations in Practical Implementation

Despite its advantages, the proposed framework has certain limitations when considered from a practical implementation perspective. The effectiveness of the approach depends on the availability and quality of input data, as well as the robustness of feature engineering and model calibration.

Additionally, the inventory simulation is based on a fixed base-stock policy, which may not fully capture the dynamic nature of real-world inventory systems. Factors such as variable lead times, supplier constraints, and external disruptions are not explicitly incorporated into the current model.

These limitations suggest that further refinement is required to enhance the applicability of the framework in operational settings.

6.9 Summary

This chapter has interpreted the results of the forecasting and inventory analysis from a practical perspective, highlighting their implications for retail decision-making. The findings demonstrate that effective inventory management requires a balanced consideration of forecasting accuracy, uncertainty, and policy alignment.

The central insight is that improvements in predictive performance must be accompanied by corresponding adaptations in inventory control strategies to achieve optimal operational outcomes. This integrated perspective provides a foundation for the development of more robust and effective retail inventory management systems.

Chapter 7

Conclusion and Future Work

7.1 Introduction

This chapter concludes the thesis by synthesising the key findings of the study and reflecting on their broader implications for demand forecasting and inventory management in retail environments. While the preceding chapters developed the methodology, presented empirical results, and interpreted their practical significance, the present chapter consolidates these insights into a cohesive summary.

In addition, the chapter identifies the limitations of the proposed framework and outlines potential directions for future research that could extend and enhance the applicability of the approach.

7.2 Summary of Research Objectives

The primary objective of this thesis was to develop a data-driven framework for demand forecasting and inventory optimization under conditions of limited and imperfect retail data. To achieve this objective, the study addressed several interconnected challenges, including data scarcity, absence of external demand signals, forecasting under uncertainty, and the translation of predictions into operational inventory decisions.

A structured methodology was proposed that integrates data reconstruction, feature engineering, machine learning-based forecasting, uncertainty quantification, and simulation-based evaluation. This end-to-end approach ensures that forecasting outputs are not evaluated in isolation but in terms of their contribution to real-world decision-making.

7.3 Summary of Key Findings

The empirical analysis conducted in this study yields several important findings that collectively advance the understanding of demand forecasting and inventory management in data-constrained retail settings.

Effectiveness of Data Reconstruction and External Signals. The integration of Google Trends data enabled the transformation of limited transactional records into a structured and temporally meaningful time series. This approach demonstrates that external signals can be leveraged to compensate for insufficient historical depth, thereby enhancing the quality of forecasting inputs.

Superiority of Feature-Based Forecasting Models. The evaluation results confirm that feature-augmented machine learning models provide superior predictive performance compared to traditional time-series approaches and naïve baselines. The ability to incorporate multiple sources of information allows such models to capture complex demand patterns that are not accessible through purely historical methods.

Importance of Residual Analysis and Uncertainty Modeling. The analysis of residual behaviour revealed that forecasting errors exhibit non-normal characteristics, including the presence of tail events. This finding highlights the limitations of conventional assumptions and demonstrates the value of empirical, residual-based approaches to uncertainty quantification.

Role of Category-Level Variability. Demand patterns and forecast uncertainty were found to vary significantly across product categories. High-demand categories exhibited greater variability and sensitivity to prediction errors, emphasising the need for category-specific inventory strategies.

Mismatch Between Forecast Accuracy and Operational Performance. A key insight of the study is that improvements in forecasting accuracy do not necessarily translate into improved inventory performance. The simulation results demonstrate that simpler forecasting approaches may outperform more advanced models when used within rigid inventory policies.

Need for Integrated Forecasting and Inventory Design. The findings collectively indicate that forecasting models and inventory policies must be designed in conjunction. The effectiveness of a forecasting model depends not only on its predictive capability but also on its alignment with the decision framework in which it operates.

7.4 Contributions of the Study

This thesis makes several contributions to the applied domain of retail analytics and inventory management.

First, it proposes a practical methodology for reconstructing a usable time series from sparse retail data through the integration of external demand signals. This approach provides a viable solution for data-constrained environments.

Second, it demonstrates the effectiveness of feature-based machine learning models in capturing demand dynamics when supported by appropriate feature engineering.

Third, it introduces a residual-driven approach to uncertainty modeling and safety stock estimation, moving beyond traditional distributional assumptions.

Finally, it provides a simulation-based evaluation framework that links forecasting performance to operational outcomes, enabling a more holistic assessment of decision effectiveness.

7.5 Limitations of the Study

While the proposed framework offers several advantages, it is subject to certain limitations that should be acknowledged.

The analysis is based on a dataset with limited historical depth and restricted feature availability. Although external data sources were used to enhance the dataset, the results may be sensitive to the quality and representativeness of these signals.

The inventory simulation assumes a fixed base-stock policy with constant parameters. In practice, inventory systems may incorporate dynamic policies that adjust to changing demand conditions.

Additionally, the study does not explicitly consider supply-side constraints such as lead time variability, supplier reliability, or logistical disruptions, which can significantly influence inventory outcomes.

These limitations provide important context for interpreting the results and highlight opportunities for further research.

7.6 Future Research Directions

Based on the findings of this study, several avenues for future research can be identified.

Dynamic Inventory Policies. Future work could explore the integration of adaptive inventory policies that adjust safety stock and reorder levels in response to real-time forecast uncertainty. Such approaches may better align with the behaviour of advanced forecasting models.

Incorporation of Additional External Signals. The inclusion of additional data sources, such as weather data, promotional activity, and pricing information, could further enhance forecasting accuracy and provide a more comprehensive representation of demand drivers.

Probabilistic Forecasting. Extending the framework to generate probabilistic forecasts rather than point estimates would enable more robust decision-making under uncertainty and improve risk-aware inventory planning.

Multi-Echelon Inventory Systems. Future studies could extend the analysis to multi-level supply chains, considering interactions between warehouses, distribution centers, and retail outlets.

Real-Time Implementation. The deployment of the proposed framework in a real-time operational setting would provide valuable insights into its practical feasibility and impact on business performance.

7.7 Final Remarks

This thesis has demonstrated that effective demand forecasting and inventory management require an integrated approach that combines data engineering, predictive modeling, uncertainty analysis, and decision optimization.

The central contribution of this work lies in showing that predictive accuracy, while important, is not sufficient on its own. The true value of a forecasting system emerges only when it aligns with the operational context in which it is applied.

By addressing the challenges of data scarcity and policy integration, this study provides a foundation for the development of more robust and practical decision-support systems in retail environments.

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pure.ulster.ac.uk <1%

30 Internet

www.ros.hw.ac.uk <1%